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EDUCATION

The University of Texas at Dallas — Dallas, TX

GPA: 3.72

B.S. in Computer Science, Minor in Mathematics

May 2028

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Discrete Mathematics, Computer Architecture, Linear Algebra, Probability and Statistics

RESEARCH EXPERIENCE

Undergraduate Research Assistant, UTD URAP — Dallas, TX

May 2026 - Present

- Selected for the **Undergraduate Research Assistant Program (URAP)** at UT Dallas under **Professor Wei Yang**, investigating the root causes of **non-determinism in LLM inference** across serving stacks, kernel implementations, and hardware backends
- Curated a dataset of **200+ documented root causes** of non-deterministic behavior by systematically triaging issues, commits, and pull requests across open-source **ML, inference-serving, and AI framework repositories**
- Developed a taxonomy of recurring failure modes—floating-point reduction order, kernel and backend selection, batching and scheduling variance, and sampling/seed handling—to ground reproducibility tooling and downstream mitigation strategies
- Built reproducible Python harnesses to isolate and replay divergent inference runs, converting anecdotal bug reports into labeled, verifiable evidence for the dataset

Student Researcher at UT Dallas — Dallas, TX

Feb 2025 - May 2025

- Improved astronomical anomaly detection to **0.85 ROC-AUC** and **0.72 recall** on the Galaxy10 dataset by designing and training a **DCGAN** architecture in **PyTorch** to generate synthetic training images ([poster](#))
- Boosted model generalization on rare celestial events to **0.81 ROC-AUC** by fine-tuning pre-trained **StyleGAN** weights to synthesize high-fidelity astronomical imagery
- Increased rare-event detection performance **15%** over baseline by engineering a GAN-based data-augmentation strategy with reproducible training and evaluation pipelines in Python

Research Assistant at Univ. of Missouri — Remote

Sept 2023 - Jan 2024

- Achieved **84% workflow-placement accuracy** across volunteer edge nodes by implementing **K-Means clustering** (scikit-learn, pandas, NumPy) over CPU, RAM, and storage telemetry
- Enabled scalable, repeatable testing across heterogeneous edge infrastructure by architecting a **Docker-based AWS (EC2) testbed** with multiple Linux VMs, tuning cluster sizing via **Elbow Method** to cut overhead

OPEN SOURCE

Contributor — OpenWorker (Andrew Ng) — Python, FastAPI, REST github.com/andrewyng/openworker, PR #185

- Added **4 NVIDIA NIM catalog models** to OpenWorker's provider ecosystem, exposing them behind an **OpenAI-compatible interface**
- Enabled that integration by first closing an authentication gap that let invalid API keys pass validation, using an inference-based verification probe with dedicated unit-test coverage

PROJECTS

AutoPM — 2nd Place out of 350+, HackUTD — Next.js, TypeScript, LangGraph, Gemini

November 2025

- Cut an estimated **10+ hours** of manual PM work per cycle by architecting an AI copilot orchestrating **8 LangGraph agents** across a 7-step workflow (research, user stories, RICE prioritization, wireframing, Jira tickets)
- Collapsed **5+ manual handoffs** into one automated pipeline by engineering type-safe agent communication with **Zod-validated** structured outputs across 3 third-party REST APIs (Jira, Gmail, Gemini)
- Delivered **7 distinct AI-generated artifacts** per project workspace by building a full-stack application in **Next.js** and React 19 with **Auth0** authentication

NetGainNBA (net-gain-nba.vercel.app) — React, Flask, Python, XGBoost, scikit-learn

February 2025 - May 2025

- Achieved **88% playoff-prediction accuracy** on 2020–2024 data by building a full-stack platform powered by a **7-model ML ensemble** (XGBoost, TabNet, Random Forest, Logistic Regression) across **89 features**
- Enabled live win-probability tracking across the playoffs by engineering an end-to-end pipeline from **NBA Stats API** ingestion through **K-Means player clustering** (9 archetypes) to ensemble inference

- Optimized load time across **26 datasets** by designing an interactive GM toolkit (drag-and-drop bracket simulation, salary-cap tracking, trade engine) with lazy loading and **Vite chunk splitting**

EduTube — *Google Cloud, React, TypeScript*

September 2025

- Built an AI-powered lecture companion integrating **TwelveLabs API** for semantic video search and **Gemini API** for auto-generated study materials, processing **100+ educational videos**
- Engineered **Fastify backend** with RESTful endpoints and **Google Cloud Storage**, implementing webhook-based real-time video processing with **95% uptime**
- Delivered responsive **React + Vite frontend** with **Framer Motion** animations, organized as type-safe monorepo architecture

UTD Spots — *Swift*

May 2025

- Developed iOS app enabling **100+ students** to discover campus study locations with amenity filtering and distance-based recommendations using **Core Location**
- Implemented **MVVM architecture** with clean separation of UI, business logic, and service layers for maintainable, testable codebase
- Integrated **MapKit** for interactive campus visualization with location-aware spot discovery and turn-by-turn navigation

LEADERSHIP AND INVOLVEMENT

AI51 Innovation Labs — *Lead Developer*

August 2025 - Present

- Leading development of **Research Mate**, an AI-powered research paper copilot featuring section-wise summarization, citation-aware explanations with source attribution, and interactive citation graph visualization
- Architected fine-tuned paper-specific models with adaptive explanation levels (professor/student/child), user profile-based recommendations, and cross-domain paper discovery using semantic similarity
- Built collaborative research environment with author analytics, FAQ autocomplete from usage patterns, hub-based communities, and expert matchmaking connecting researchers across experience levels

TuneTrend, UT Dallas AI Mentorship Program (AIM) — *Product Manager*

August 2025 - Present

- Directed a **4–5 person engineering team** to ship a genre-aware music popularity predictor by owning the architecture and an ML pipeline extracting **140+ audio features** (MFCCs, chroma, spectral contrast) with artist-level **GroupShuffleSplit**
- Shipped a full-stack analysis platform serving **6 genre models** by building real-time audio hook detection, 3D visualizations, and nearest-neighbor track matching in **Next.js, Flask, and Three.js**

Artificial Intelligence Society (AIS) — *Officer*

May 2025 - Present

- Organizing **HackAI**, UTD's premier annual AI hackathon, managing event logistics, corporate sponsorships, and technical challenges for hundreds of participants
- Facilitating technical workshops and community initiatives to promote AI literacy and hands-on project development within the student body

HONORS AND AWARDS

- **2nd Place, PNC Track — HackUTD 2025** (out of 350+ participants)

November 2025

- **Dean's List — Fall 2024**

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, Swift, HTML/CSS

AI/ML & Data: PyTorch, scikit-learn, XGBoost, pandas, NumPy, LangChain, LangGraph, GANs (DCGAN, StyleGAN), K-Means, librosa

Web & Backend: React, Next.js, Node.js, Flask, Spring Boot, Fastify, REST APIs, MongoDB, PostgreSQL, Auth0, Zod

Cloud & DevOps: AWS (EC2, S3), Google Cloud Storage, Docker, Kubernetes, MicroK8s, OpenFaaS, Linux/Unix, Git, CI/CD

Tools & Practices: Agile/Scrum, Unit Testing, Jira, Postman, Streamlit, Vite, Gmail API, Gemini API